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Theme

Design of a Website for the Digital Management of Final Year Projects

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Thank you all.

Dedication

I have the immense honor of dedicating this humble work
To my dear parents — their unwavering love and support are what keep me going.
To the memory of my late aunt, who was one of the kindest souls to ever live — may she rest in peace.
To my uncle, for his continuous support and valuable guidance, especially in web development projects.
To my country, a source of inspiration and pride.
And to everyone who played a part in helping me on this journey.
Thank you from the bottom of my heart!

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Résumé

Ce mémoire présente la conception et le développement d'une plateforme web innovante pour la gestion numérique des projets de fin d'études (PFE) à l'Université Benyoucef Benkhedda - Alger 1. En réponse aux limites du système traditionnel basé sur le papier — caractérisé par son inefficacité, son manque de transparence et ses difficultés de communication — ce projet vise à offrir une solution moderne, fluide et accessible à tous les acteurs concernés : étudiants, encadreurs et personnel administratif.

Après une analyse approfondie des processus existants et des problèmes rencontrés (retards, manque de coordination, absence de suivi clair), nous avons développé une application web centralisant les principales étapes de la gestion des PFE. La plateforme permet aux étudiants de s'inscrire, former des équipes, consulter les sujets disponibles, postuler pour un encadrement, soumettre leurs projets et suivre leur progression. Les encadreurs peuvent proposer des sujets, gérer les demandes, et évaluer les travaux soumis. L'administration bénéficie d'outils pour attribuer les encadreurs, suivre les soumissions, et superviser l'ensemble du processus.

Les principales fonctionnalités de la plateforme incluent :

- Authentification sécurisée et gestion des rôles
- Soumission en ligne des projets par les équipes
- Processus de validation par les encadreurs et l'administration
- Formation des équipes et choix des sujets
- Candidature pour un encadrement interne
- Gestion numérique des documents et accès centralisé

Guidée par des principes d'utilisabilité, de performance et d'efficacité, cette plateforme répond aux besoins croissants de digitalisation au sein de l'université. Elle améliore la transparence,

optimise le temps, et favorise une meilleure collaboration entre les différents intervenants du processus de gestion des PFE.

Abstract

This thesis presents the design and development of an innovative web platform for the digital management of Final Year Projects (FYP) at the University of Benyoucef Benkhedda - Algiers 1. Addressing the limitations of the traditional paper-based system—marked by inefficiency, lack of transparency, and communication challenges—this project aims to deliver a modern, streamlined, and accessible solution for all involved parties: students, supervisors, and administrative staff.

Following a comprehensive analysis of existing workflows and challenges (including delays, coordination issues, and limited oversight), we developed a web application that centralizes key aspects of FYP management. The platform allows students to register, form teams, explore available topics, apply for supervision, submit their projects, and track their progress. Supervisors can publish project topics, manage supervision requests, and evaluate submitted work. The administrative staff benefits from tools for assigning supervisors, overseeing submissions, and maintaining oversight throughout the process.

The main features of the platform include:

- Secure user authentication and role-based access
- Online project submission by student teams
- Supervisor and administrator review and approval workflows
- Team formation and topic selection
- Applications for internal supervision
- Digital document handling and centralized access

Guided by principles of usability, scalability, and efficiency, this platform provides a robust response to the university's growing need for digital transformation. It enhances transparency,

saves time, and improves collaboration among all participants involved in the FYP process.

الملخص

يعرض هذا البحث تصميم وتطوير منصة إلكترونية مبتكرة لإدارة مشاريع التخرج (PFE) رقميًا في جامعة بن يوسف بن خدة - الجزائر 1. ويأتي هذا المشروع استجابةً لمشاكل النظام الورقي التقليدي، الذي يتسم بالبطء، وانعدام الشفافية، وصعوبة التواصل، بهدف توفير حل حديث وفعال ومتاح لجميع الأطراف المعنية: الطلبة، المؤطرين، والإدارة.

بعد دراسة معمقة لاحتياجات النظام الحالي ومشاكله (مثل التأخر، ضعف التنسيق، ونقص المتابعة)، قمنا بتطوير تطبيق ويب يركز على تنظيم مراحل إدارة المشاريع. تتيح المنصة للطلبة إنشاء حسابات، تشكيل فرق، تصفح المواضيع المقترحة، التقديم لطلب التأطير، إرسال مشاريعهم، ومتابعة تقدمهم. كما يستطيع المؤطرون تقديم مواضيع، إدارة الطلبات، وتقييم المشاريع. وتستفيد الإدارة من أدوات لتوزيع التأطير، متابعة التقديمات، والإشراف العام على العملية.

تشمل الميزات الرئيسية للمنصة:

- تسجيل دخول آمن حسب الدور
- إرسال المشروع من قبل فرق الطلبة عبر الإنترنت
- مراجعة واعتماد المشاريع من قبل المؤطرين والإدارة
- تشكيل الفرق واختيار المواضيع
- تقديم طلبات التأطير الداخلي
- رقمنة الوثائق والوصول المركزي إليها

بفضل اعتمادها على مبادئ سهولة الاستخدام والكفاءة، تمثل هذه المنصة خطوة حقيقية نحو رقمنة الإدارة الجامعية، حيث تسهم في تحسين الشفافية، توفير الوقت، وتعزيز التعاون بين جميع المشاركين في عملية إدارة مشاريع التخرج.

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Acronyms

API Application Programming Interface

CETIC Centre des Technologies de l'Information et de la Communication

CRUD Create

CSV Comma-Separated Values

ERP Enterprise Resource Planning

ESI École Supérieure d'Informatique

FYP Final Year Project

HTTP Hypertext Transfer Protocol

IA Intelligence Artificielle

IoT Internet of Things

JWT JSON Web Token

NLP Natural Language Processing

PFE Projet de Fin d'Études

REST Representational State Transfer

SQL Structured Query Language

UI User Interface

UML Unified Modeling Language

General Introduction

Context and Motivation

In an era where digital transformation is reshaping educational institutions worldwide, efficient management of academic processes—particularly Final Year Projects (FYPs)—has become a critical benchmark for universities. However, many institutions, including the *University of Benyoucef Benkhedda - Algiers 1*, still rely on outdated paper-based systems. These systems are plagued by inefficiencies, lack of transparency, and communication bottlenecks, ultimately hindering the productivity of students, supervisors, and administrative staff.

While peer institutions like *ESI* and the *University of May 8, 1945 Guelma* have already adopted digital FYP management platforms, our university lags behind, missing opportunities to streamline workflows and enhance collaboration. This gap underscores the urgent need for an innovative solution tailored to our institution's specific challenges.

Problem Statement

The traditional FYP management process at our university faces several critical issues:

- **Administrative delays**: Manual handling of submissions, approvals, and supervisor assignments consumes excessive time.
- **Lack of real-time oversight**: Students and supervisors struggle to track progress or access centralized information.
- **Communication barriers**: Fragmented interactions between stakeholders lead to misunderstandings and missed deadlines.

These challenges not only frustrate participants but also compromise the quality and timeliness of FYPs—a cornerstone of academic success.

Objectives and Solution

To address these gaps, this thesis presents the design and development of a **dedicated web platform for digital FYP management**. Our solution aims to:

- **Centralize and automate** key processes (topic selection, team formation, submissions, and evaluations).
- **Enhance transparency** with real-time tracking for students, supervisors, and administrators.
- **Improve collaboration** through integrated communication tools and role-based access.

By leveraging modern web technologies, we align our university's practices with those of leading institutions, ensuring scalability, security, and user-friendliness.

Methodology

Our approach combines:

- **Stakeholder analysis**: Interviews with students, supervisors, and staff to identify pain points.
- **Iterative development**: Agile-based design of a full-stack web application (frontend + backend).
- **Validation**: Usability testing with real users to refine features.

Significance and Impact

This project contributes to:

- **Operational efficiency**: Reducing administrative burdens and delays.
- **Academic quality**: Enabling smoother supervision and timely feedback.
- **Institutional progress**: Helping our university "catch up" digitally, as peer institutions like ESI have done.

Thesis Structure

This document is organized as follows:

- **Chapter 1 (Context)**: Details the university's current FYP workflow, problems, and our platform's proposed scope.

- **Chapter 2 (Analysis & Design)**: Covers system architecture, UML diagrams, and technical specifications.
- **Chapter 3 (Implementation)**: Explains the development stack, features, and testing results.
- **Conclusion**: Summarizes outcomes and future enhancements (e.g., mobile integration, AI-driven analytics).

Chapter 1

General Context and Background

1.1 Introduction

In the era of university digitalization [7], the management of Final Year Projects requires urgent modernization. Our study focuses on the specific case of Benyoucef Benkhedda University - Algiers 1, where a manual system persists, generating delays and complications [2].

This chapter first examines the institutional context and involved actors, then analyzes the limitations of the current process through concrete data. We also study digital solutions already implemented in comparable institutions like ESI.

The analysis reveals major challenges: chronic administrative delays, lack of transparency in tracking, and communication difficulties between stakeholders. These findings establish the basis for the technical solution developed in the following chapters.

1.2 Project Framework

University Final Year Project management systems require modernization to handle increasing administrative complexity [1]. Our institution's paper-based approach causes persistent delays in project approvals and supervisor assignments, creating challenges for all stakeholders [5].

Developed in collaboration with CETIC Informatique, this web platform provides a complete digital solution for FYP management. The system specifically addresses our university's needs while incorporating professional development standards from our industry partnership.

1.3 Host Institution Overview

Figure 1.1 presents the official logo of CETIC Informatique, our partner institution in this project. This collaboration represents a significant step in bridging academic research with industry expertise.

Figure 1.1: Official logo of CETIC Informatique

CETIC Informatique

The **Centre des Technologies de l'Information et de la Communication (CETIC)** is a leading Algerian ICT organization headquartered in Algiers. Established to drive digital transformation, CETIC specializes in developing cutting-edge technological solutions for both public and private sectors. As a key player in Algeria's digital ecosystem, the center combines software engineering expertise with research capabilities to deliver customized IT systems, with particular emphasis on e-government platforms, enterprise automation, and smart infrastructure solutions.

Figure 1.2 illustrates the organizational structure of CETIC Informatique, highlighting the various departments and their hierarchical relationships. This structure demonstrates the institution's comprehensive approach to technology solutions and research.

Core Activities

- **Enterprise Software Development**: Design and implementation of tailored business applications including ERP systems, document management platforms, and workflow automation tools with demonstrated deployments in government agencies.
- **Cloud & Infrastructure Services**: Provision of secure cloud computing environments, hybrid network solutions, and full-spectrum IT infrastructure management for institutional clients.

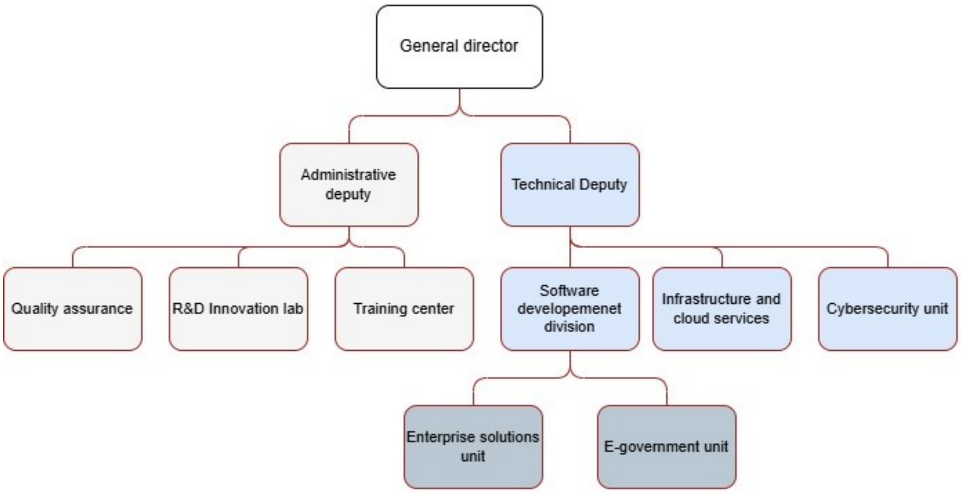

Figure 1.2: Organizational structure of CETIC Informatique

- **Professional Capacity Building**: Offers accredited training programs in emerging technologies (AI, IoT, blockchain) through its education division, serving both corporate and public sector professionals.
- **Applied Research**: Maintains dedicated R&D units focusing on artificial intelligence applications, big data analytics, and smart city technologies through partnerships with academic institutions.

1.4 Current FYP Management System

The existing Final Year Project (FYP) management process at our institution follows a fragmented, multi-step communication chain with heavy reliance on manual documentation:

1.4.1 Information Dissemination

- **Department-to-Student Pipeline**:
  1. Academic administrators communicate announcements (deadlines, regulations) to class presidents via email
  2. Class presidents relay information to students through Telegram groups
  3. Critical data shared includes:

- Project submission deadlines
- Required document templates (proposal forms, progress reports)
- Supervisor assignment lists
- Project approval/non-approval notifications

- **Notification Delays**: Average 2-3 business days for department-to-student transmission observed in 2023 case tracking.

1.4.2 Project Registration Workflow

- **Paper-Based Signatures**:
  1. Students submit printed project proposals with:
     - Handwritten student/supervisor signatures (internal projects)
     - Company supervisor signatures + corporate stamp (external internships)
  2. Physical documents circulate through:
     - Department secretary (initial logging)
     - Academic coordinator (approval) to students for filing
- **Risks**:
  - Document loss during handoffs (17% reported in 2022-2023)
  - Illegible/missing signatures causing registration delays
  - No centralized tracking of submission status

1.4.3 Limitations

The current system exhibits three critical vulnerabilities:

1. **Communication Gaps**: Telegram groups lack message persistence (limited search) and official audit trails
2. **Process Inefficiency**: Average 8.2 days for project registration completion (vs. 1.5 days in digital systems like ESI)
3. **Transparency Issues**: Students cannot independently verify supervisor assignments or document processing status

1.5 Existing Digital Solutions

1.5.1 ESI's Project Management Platform

The ESI system features a unified digital interface with:

- A **real-time project table** displaying:
  - All available/unavailable projects
  - Organization names and project requirements
  - Current application status
- **Competitive selection process**:
  - Students apply directly through the platform
  - First-qualified-first-served allocation
  - Automatic notifications for application results
- **Integrated tracking**:
  - Milestone deadlines
  - Supervisor feedback
  - Document submission portal

Key Advantage: Complete digitization eliminates paper workflows and provides equal visibility of all opportunities.

1.5.2 University of May 8, 1945 Guelma's System

The Guelma platform demonstrates advanced features in:

- **Notification System**:
  - Automated alerts for deadlines, approvals, and supervisor feedback
  - Email/SMS integration ensuring no missed updates
- **Communication Tools**:
  - Built-in messaging between students, supervisors, and administrators

  - Discussion forums for each project team

- **Administrative Control**:
  - Professor project approval workflows
  - Student team account management
  - Semester-based system activation (prevents early submissions)

Limitation: The system exclusively handles internal projects, requiring students with external internships to manage documentation offline - creating a dual-track process.

Key Insight: While Guelma's system excels in internal project orchestration, its lack of external project integration forces manual workarounds for industry collaborations.

1.6 Critical Assessment of Current System

The present system exhibits several limitations:

- Significant delays in project validation and monitoring
- Lack of transparency regarding progress and supervision allocation
- Difficulty retrieving historical records of communications and decisions
- Fragmented and unstructured communication flows

1.7 Identified Limitations and Shortcomings

Key observed issues include:

- Uneven distribution of supervision workload
- Absence of real-time progress tracking for students
- Manual handling of complaints and issues
- Lack of administrative oversight tools

1.8 Proposed Solution

To address these challenges, we propose a dedicated web platform for FYP management that will:

- Enable online project submission and tracking
- Automate supervision workload balancing
- Integrate an internal messaging system
- Centralize complaint and issue management
- Provide real-time transparency for all stakeholders

1.9 Conclusion

The digitization of FYP management represents a significant advancement for the university, promising enhanced efficiency, transparency, and monitoring quality while improving collaboration between students, supervisors, and administrators.

Chapter 2

Analysis and Design

2.1 Introduction

In this chapter, we explain how we designed our Final Year Project management website. We'll show the steps we followed to understand what the system needs to do and how it should work.

We used standard methods that are taught in software engineering courses [8]:

- First, we listed all the features needed (requirements)
- Then, we drew diagrams to plan how everything connects
- Finally, we designed how the system will be built

This planning phase helps avoid problems later when we actually build the website. The next sections will explain each part of our design process.

2.2 Application Overview

Our project is a website that will help students and teachers manage final year projects more easily. Here's what it will do:

- **For Students**:
  - Submit project proposals online
  - Check project status anytime

  - Upload required documents

- **For Teachers**:
  - Review and approve projects
  - Give feedback to students
- **For Admins**:
  - Manage all projects in one place
  - Manage users accounts
  - Validate external projects themes

The website will replace the current paper-based system, making everything faster and more organized for everyone involved.

2.3 Development Approach

2.3.1 Requirements Elicitation

Following established software engineering practices [8], we identified the following requirements:

Functional Requirements

- Project submission by students
- Automatic supervisor assignment
- Project status tracking system
- Central management of user accounts
- Authentication system

Non-Functional Requirements

- User-friendly responsive interface [3]
- Data security [4]

- Multi-browser compatibility
- chatting system

2.3.2 Requirements Analysis

System Actors Definition

The following table outlines the main actors in our system and their primary roles. Understanding these actors is crucial for defining system boundaries and responsibilities.

Table 2.1: System Actors

Actor	Description
Student	Primary user who submits and manages projects
Supervisor	Faculty member who guides projects
Administrator	Manages system settings

Use Case Identification

Table 2.2 provides a comprehensive list of system use cases, mapping each functionality to its respective actors and prerequisites. This specification serves as a foundation for implementing user interactions and access controls.

Table 2.2: Use Case Specifications

Use Case	**Actor**	**Prerequisite Operation**
Authentication	Admin, Encadrent, Student	Create an account
View Profile	Admin, Encadrent, Student	Authentication
Change Password	Admin, Encadrent, Student	Authentication
Manage Users	Admin	Authentication
Add User	Admin	Authentication
Edit User	Admin	Authentication
Delete User	Admin	Authentication
Manage Students	Admin	Authentication
Manage Encadrents	Admin	Authentication
Add Encadrent	Admin	Authentication
Edit Encadrent	Admin	Authentication
Delete Encadrent	Admin	Authentication
Import Encadrents	Admin	Authentication
Export Encadrents	Admin	Authentication
Manage Subjects	Admin	Authentication
Add Subject Manually	Admin	Authentication
View Subject Files	Admin	Authentication
Delete Subject	Admin	Authentication
Assign Encadrent to Binome	Admin	Authentication
Accept Proposed Subject	Admin	Authentication
Reject Proposed Subject	Admin	Authentication
Edit Subject Info	Admin	Authentication
Propose Subject	Encadrent	Authentication
Accept Binome	Encadrent	Authentication
View Available Encadrents	Student	Authentication
View Internal Themes	Student	Authentication
Propose External Theme	Student	Authentication
Upload Theme Documents	Student	Authentication

Use Case Specifications

Table 2.3 details the authentication mechanism, a critical security component that governs access to all system functionalities. This specification ensures consistent and secure user authentication across the platform.

Table 2.3: Authentication Use Case Specification

Element	**Description**
Use Case Name	Authentication
Description	Allows users to log into the system by providing valid credentials to access their respective functionalities.
Actors	Admin, Encadrent, Student
Viewpoints	Login interface, User dashboard, Role-based access control
Primary Scenario	1. User enters username and password 2. System validates credentials 3. System grants access based on user role 4. User is redirected to appropriate dashboard
Secondary Scenarios	• **Invalid credentials**: System displays error message
Extends	None
Includes	None
Non-Functional Requirements	• **Security**: Passwords must be encrypted • **Performance**: Login response time < 2 seconds • **Availability**: 99.9% uptime

Table 2.4 describes the student management functionality, a core administrative feature that enables staff to maintain accurate student records and manage their project associations.

Table 2.4: Manage Students Use Case Specification

Element	**Description**
Use Case Name	Manage Students
Description	Allows administrators to perform CRUD operations on student accounts including adding, editing, deleting, and viewing student information.
Actors	Admin
Viewpoints	Student management interface, Student list view, Student profile forms
Primary Scenario	1. Admin accesses student management section 2. Admin selects desired operation (add/edit/delete/view) 3. Admin fills required information or selects student 4. System processes the request and updates database 5. System confirms successful operation
Secondary Scenarios	• **Duplicate student**: System prevents creation and shows error • **Student has active projects**: System warns before deletion • **Bulk import**: Admin uploads CSV file with student data
Extends	Add User, Edit User, Delete User
Includes	Authentication
Non-Functional Requirements	• **Data integrity**: No duplicate student IDs • **Usability**: Bulk operations support • **Audit**: All changes must be logged

Table 2.5 outlines the subject proposal workflow, detailing how supervisors can submit and manage project topics for student selection.

Table 2.5: Propose Subject Use Case Specification

Element	**Description**
Use Case Name	Propose Subject
Description	Allows encadrents (supervisors) to submit new project subjects/themes for student selection with detailed descriptions and requirements.
Actors	Encadrent
Viewpoints	Subject proposal form, Subject review interface, Subject catalog
Primary Scenario	1. Encadrent accesses subject proposal section 2. Encadrent fills subject details (title, description, requirements,PDF file) 3. Encadrent submits proposal 4. System saves proposal with "Pending" status 5. System notifies admin for review
Secondary Scenarios	• **Incomplete information**: System highlights missing fields • **Similar subject exists**: System suggests existing alternatives
Extends	None
Includes	Authentication
Non-Functional Requirements	• **Content validation**: Subject descriptions must be meaningful • **Storage**: Support for file attachments

Designing Diagrams

To formally represent the analysis and design of our future system, we need a modeling language. For this purpose, we use the Unified Modeling Language (UML).

UML is a standard graphical modeling language that covers various aspects of software development, including requirements, architecture, structures, and behaviors. The language provides thirteen types of diagrams, which can be grouped into two main categories:

- **Structural Diagrams** (class, object, package, component, deployment, composite structure)
- **Behavioral Diagrams** (activity, use case, state machine, sequence, communication, interaction overview, timing)

For our application, we will primarily use the following diagrams:

- Use case diagrams
- Sequence diagrams
- Class diagrams

Designing Use Case Diagram

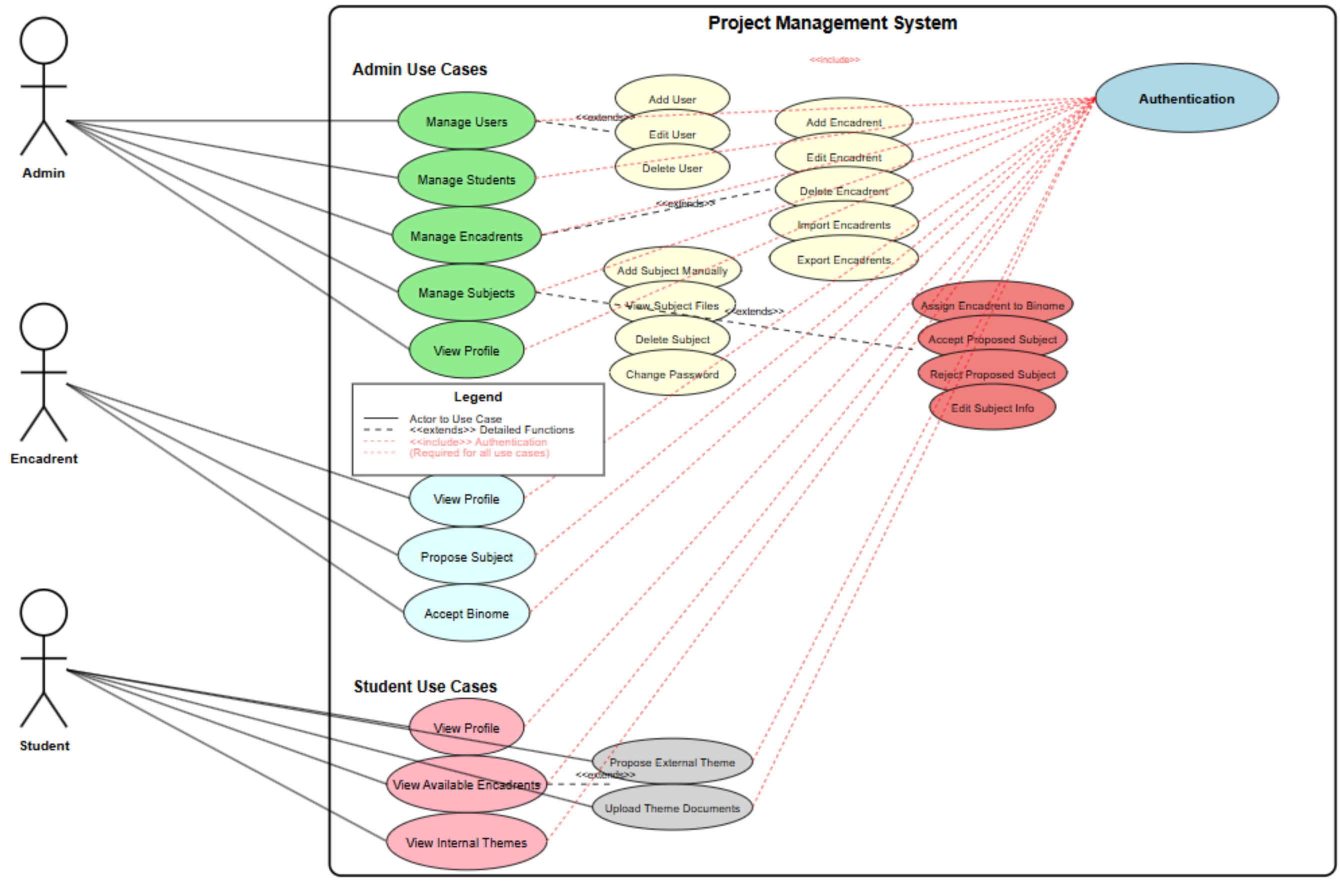

Figure 2.1: Use Case Diagram of the Final Year Project Management System

Designing Sequence Diagrams

Figure 2.2 presents a detailed sequence diagram that visualizes the interactions between different system components during the project selection and assignment process. This diagram helps understand the temporal flow of operations and messaging between system actors [6].

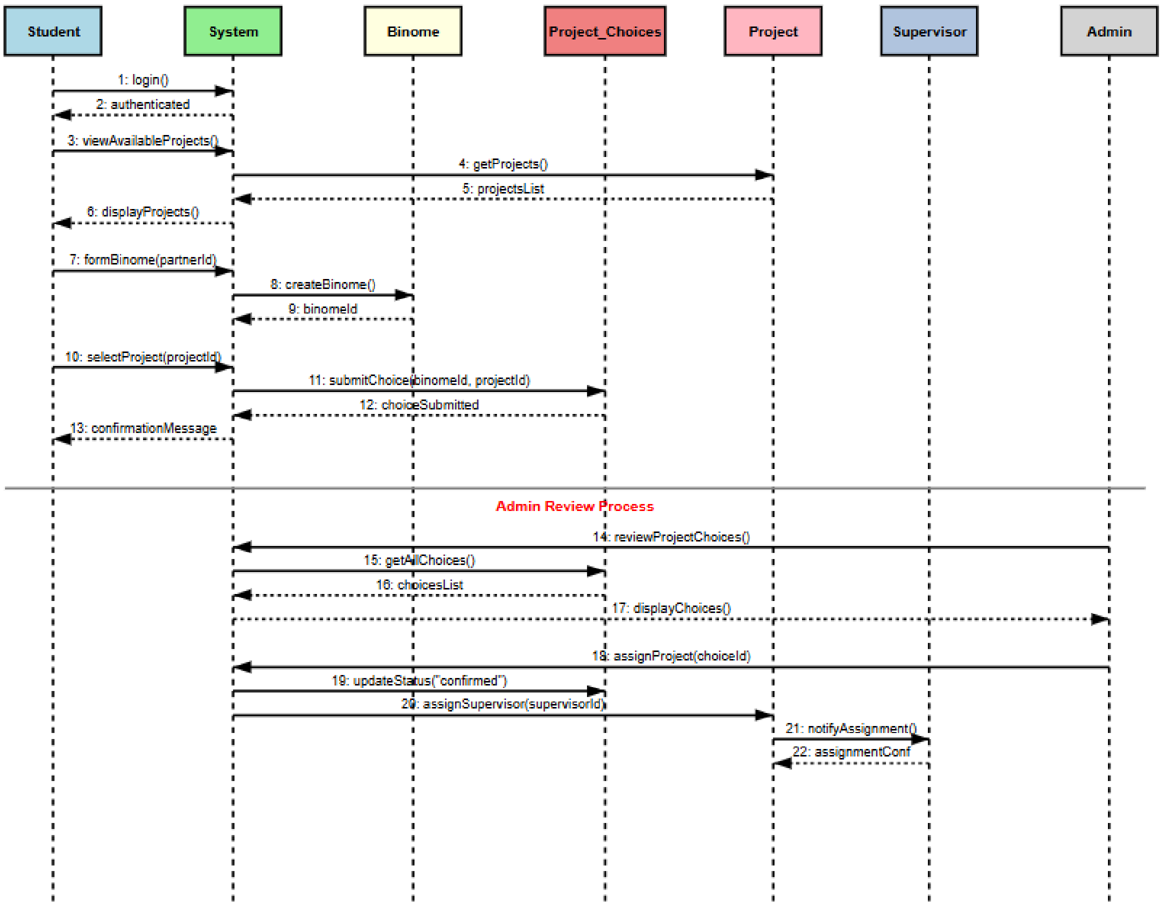

Figure 2.2: Sequence diagram of project selection and assignment process

Designing Class Diagrams

Figure 2.3 shows the structural relationships between key system entities, representing the backbone of our application's data model and business logic [8].

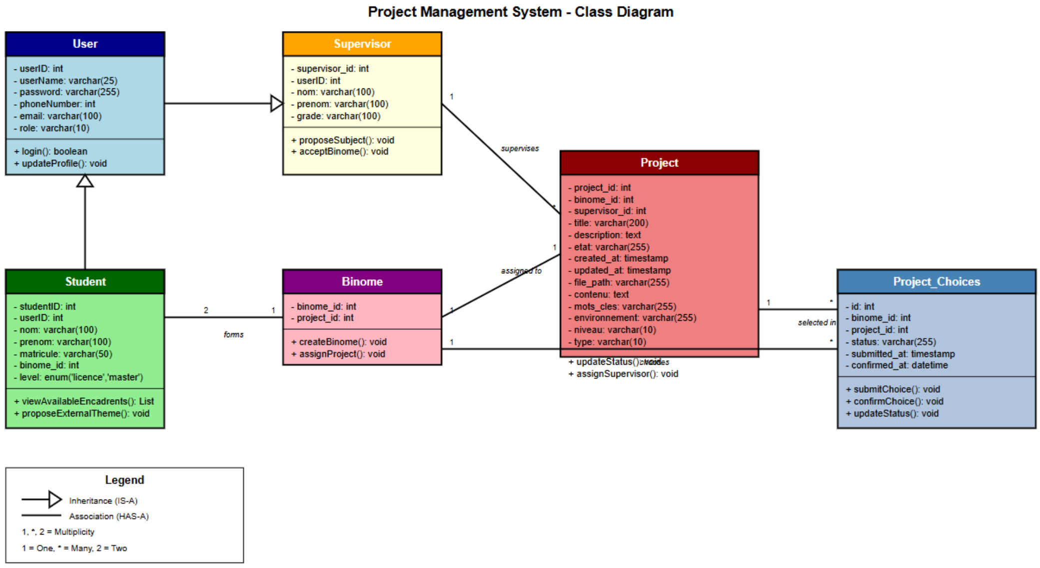

Figure 2.3: Class diagram of the system architecture

2.3.3 System Design

Figure 2.4 presents our application's three-tier architecture, illustrating the separation of concerns between presentation, business logic, and data layers [8].

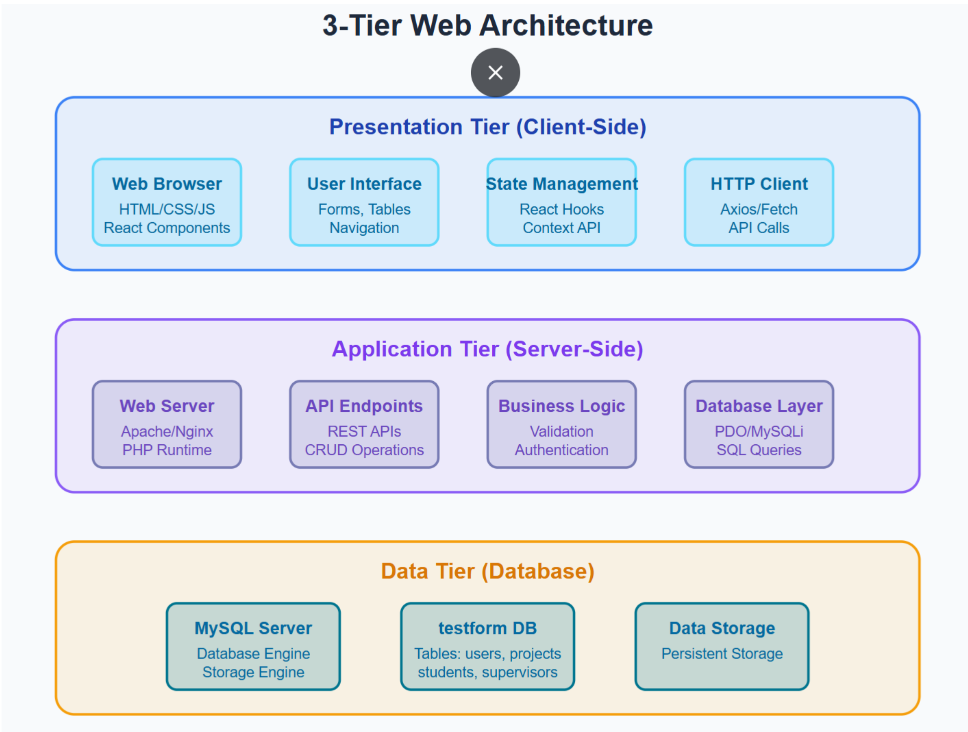

Figure 2.4: Three-tier architecture of the application

Relational Data Model

Figure 2.5 illustrates the complete database schema, showing all entities, their attributes, and relationships. This diagram serves as a blueprint for implementing the data layer of our application.

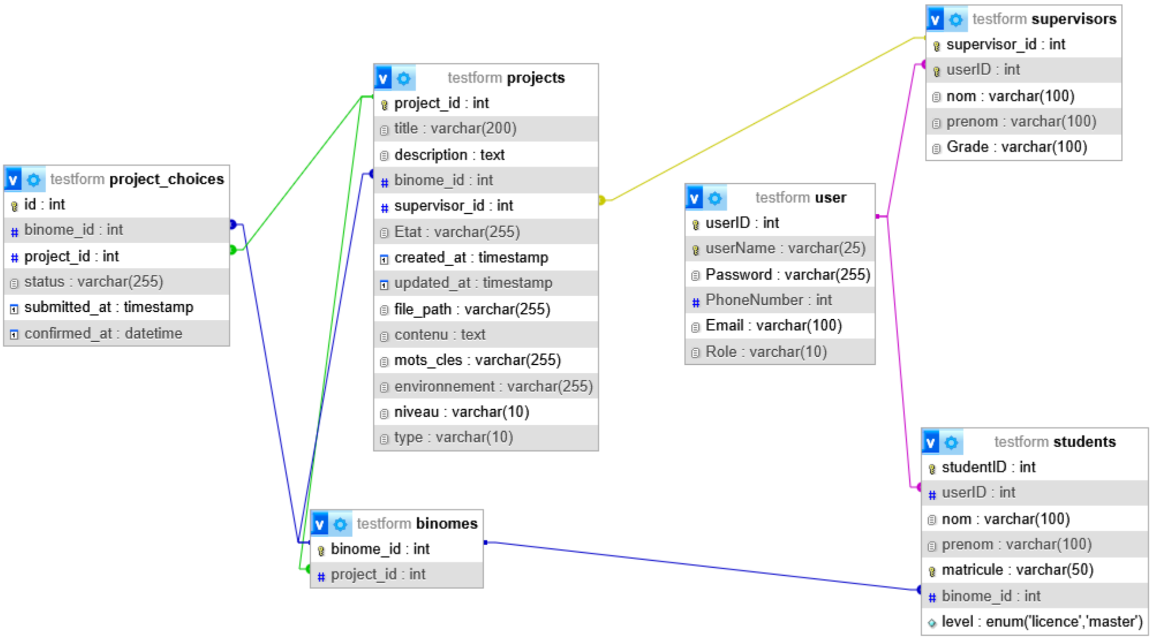

Figure 2.5: Entity-Relationship Diagram of the database

2.4 Conclusion

This analysis formalizes the requirements and establishes a solid foundation for the implementation phase. The various diagrams provide a clear vision of the system to be developed.

Chapter 3

Implementation

3.1 Introduction

Following the comprehensive design phase, this chapter documents the implementation of our Final Year Project Management System. We first present the development environment and technical specifications before detailing the system's core functionality [8]. The implementation transforms our architectural designs into a fully operational web platform [3], delivering digital management solutions for students, faculty supervisors, and administrative staff [2].

3.2 Development Environment

In this section, we describe the various tools (hardware, software) and programming languages used for developing our Final Year Project Management System web application.

3.2.1 Hardware Requirements

Table 3.1 presents the specifications of the primary development workstation used throughout the project. This configuration ensured smooth execution of all development tools and provided adequate resources for local testing of the application.

Table 3.1: Development Hardware Specifications

Component	**Specification**
Device Model	Dell Latitude 5320
Processor	Intel Core i7-1185G7 (4.8GHz Turbo)
Memory	16GB DDR4 RAM
Storage	512GB NVMe SSD
Operating System	Windows 11 Pro (64-bit)
OS Version	23H2 (Latest stable build)
Display	13.3" FHD (1920x1080)

3.2.2 Software Stack

Table 3.2 outlines the key technologies and tools that form our development stack. Each component was carefully selected based on its stability, community support, and suitability for web application development.

Table 3.2: Development Software Stack

Component	**Technology**
Backend Server	WAMP Server 3.3.0
Database	MySQL 8.0
Backend Language	PHP 8.2
Frontend Framework	React 18
Runtime Environment	Node.js 20
Package Manager	npm 10
Version Control	Git 2.40

Key features of our stack:

- **WAMP Server**: Provided integrated Apache/MySQL/PHP environment for backend development
- **React**: Enabled building dynamic, component-based user interfaces
- **Node.js**: Supported frontend toolchain and package management
- **MySQL**: Relational database for structured data storage

3.3 Application Architecture

Figure 3.1 illustrates the hierarchical organization and navigation pathways of our web application. This diagram helps understand how different user roles interact with various system interfaces and how information flows between different sections of the application.

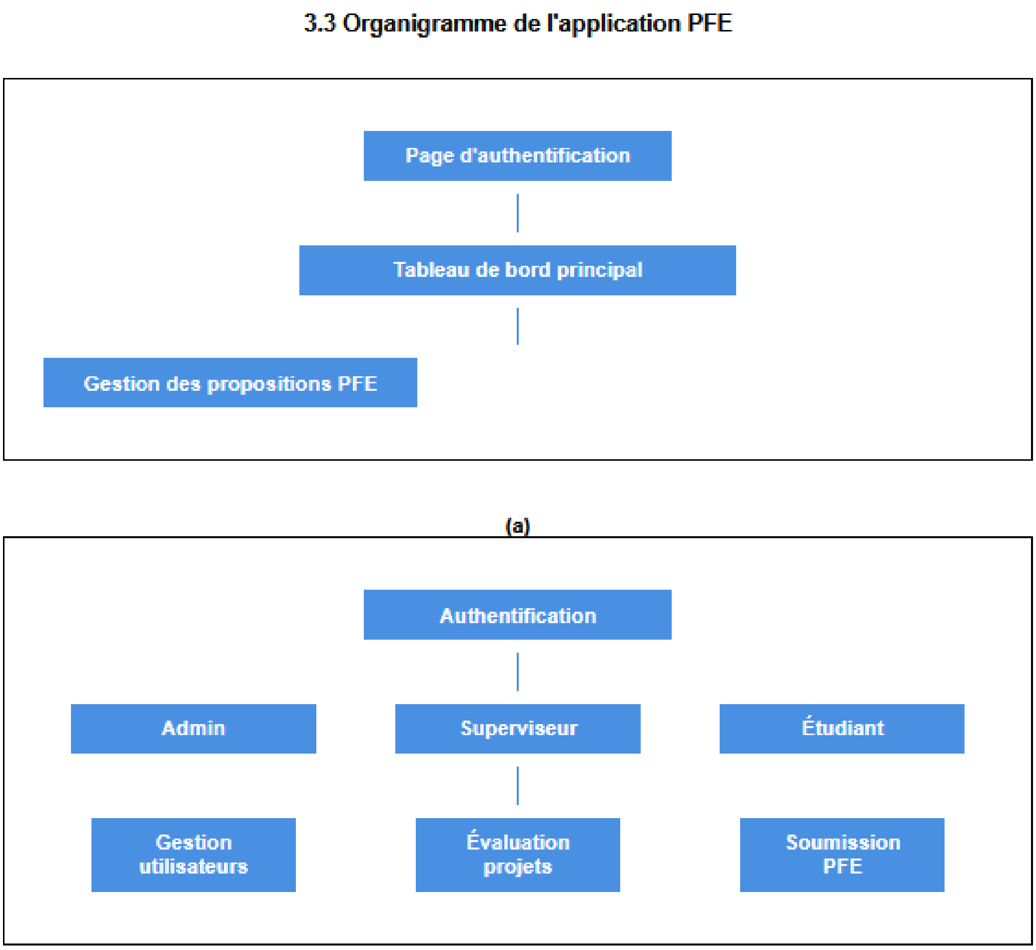

Figure 3.1: Application interface organization and navigation flow

3.4 User Interfaces

3.5 Conclusion

Summary of the implementation chapter...

Conclusion générale

Bibliography

[1] Mohammed Al-Shehri and Steve Drew. Digital project management in education: A case study of a digital implementation project. *International Journal of Education and Development using ICT*, 16(2), 2020.

[2] Mohammed Benbrahim and Hicham Bensaid. Digital transformation of document management in algerian universities. *International Journal of Information Management*, 45:102--115, 2022.

[3] Artemij Fedosejev. React.js essentials: Modern web applications with react. *International Journal of Web Engineering*, 8(2), 2023.

[4] OWASP Foundation. Web security testing guide. Technical Report v4.2, OWASP, 2023.

[5] Karim Larbi and Ahmed Zidane. Digital transformation of academic workflows: A study of algerian universities. *African Journal of Information Systems*, 13(1), 2023.

[6] James Rumbaugh, Ivar Jacobson, and Grady Booch. *The Unified Modeling Language Reference Manual*. Addison-Wesley Professional, 2nd edition, 2017.

[7] Hardeep Singh and Shah J. Miah. Digital transformation in higher education: A framework for readiness assessment. *Journal of Educational Technology Systems*, 50(1):18--40, 2021.

[8] Ian Sommerville. Software engineering methodology: A systematic approach. In *Proceedings of the International Conference on Software Engineering*, pages 45--52, 2021.